

Positive Square Energy of Theta Graphs

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Abstract

For a graph G with adjacency eigenvalues λ_i , let $s^+(G) = \sum_{\lambda_i > 0} \lambda_i^2$. We prove that every simple theta graph G satisfies $s^+(G) > |V(G)|$. This verifies the theta-graph case of the conjecture of Akbari, Kumar, Mohar, Pragada, and Zhang [?] that every connected n -vertex graph with at least $n + 1$ edges has $s^+ \geq n$. The proof combines bipartite spectral symmetry, an improved induced- P_3 removal lemma, and explicit positive-semidefinite variational witnesses for the three short exceptional parity families.

1 Introduction

A simple theta graph $\Theta(a, b, c)$ consists of three internally vertex-disjoint paths of positive integer edge lengths a, b, c between the same pair of branch vertices; at most one of a, b, c is one. It has

$$n = a + b + c - 1, \quad m = a + b + c = n + 1.$$

Our result is the following.

Theorem 1. *Every simple theta graph G satisfies $s^+(G) > |V(G)|$.*

We use two standard facts. If G is bipartite, its adjacency spectrum is symmetric; together with $\text{tr}(A(G)^2) = 2|E(G)|$, this gives

$$s^+(G) = |E(G)|. \tag{1}$$

For a real symmetric matrix M , define $s^+(M)$ analogously. Then

$$s^+(M) = \max_{X \succeq 0} \{2 \text{tr}(MX) - \text{tr}(X^2)\}. \tag{2}$$

The latter follows by diagonalizing M .

We require the following rigorous form of the improved P_3 -removal lemma of Akbari et al. [?, Lemma 2.4]. Their proof reduces the improvement to a scalar inequality checked numerically; we include an exact verification.

Lemma 2 (Improved P_3 removal). *If three vertices U of a graph G induce a P_3 , then some $v \in U$ satisfies*

$$s^+(G) \geq s^+(G - v) + \frac{17}{16}. \tag{3}$$

Proof. Put $C = 17/16$ and

$$B = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

For a symmetric matrix Z , let $r_i(Z) = Z_{ii}^2 + 2 \sum_{j \neq i} Z_{ij}^2$, the Frobenius contribution removed when row and column i are deleted. We show that for every 3×3 matrix $M \succeq 0$, some i has $r_i(B+M) > C$.

Conjugating by $\text{diag}(1, -1, 1)$ reduces this to $B - M$. Write

$$M = \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix} \succeq 0$$

and suppose all three row/column sums are at most C . Explicitly,

$$\begin{aligned} a^2 + 2(1-b)^2 + 2d^2 &\leq C, \\ c^2 + 2(1-b)^2 + 2(1-e)^2 &\leq C, \\ f^2 + 2d^2 + 2(1-e)^2 &\leq C. \end{aligned}$$

With $x = (b+e)/2$, convexity and Cauchy–Schwarz give

$$c^2 \leq C - 4(1-x)^2, \tag{4}$$

$$(a+f+2d)^2 \leq 6(C - 2(1-x)^2). \tag{5}$$

Moreover $M \succeq 0$ gives

$$c(a+f+2d) \geq (b+e)^2 = 4x^2. \tag{6}$$

The initial inequalities imply $15/32 \leq x \leq 17/16$: the lower bound follows from $|1-x| \leq \sqrt{C}/2$ and $\sqrt{C} \leq C$, while $b^2 \leq ac$, $e^2 \leq cf$, and $a, c, f \leq \sqrt{C}$ give $x \leq C$. Equations (??)–(??) would imply

$$16x^4 \leq 6(C - 4(1-x)^2)(C - 2(1-x)^2).$$

The reverse inequality holds exactly. Its left side minus right side is

$$F(x) = -32x^4 + 192x^3 - \frac{999}{4}x^2 + \frac{231}{2}x - \frac{2115}{128}.$$

Under affine parametrization of $[15/32, 2/3]$, its degree-four Bernstein coefficients are

$$\frac{31815}{32768}, \frac{5821}{8192}, \frac{2515}{8192}, \frac{37}{2304}, \frac{461}{10368};$$

on $[2/3, 17/16]$ they are

$$\frac{461}{10368}, \frac{175}{1728}, \frac{26459}{18432}, \frac{35027}{6144}, \frac{28193}{2048}.$$

All are positive, while Bernstein basis functions are nonnegative. This proves the local matrix assertion.

Finally,

$$s^+(H) = \min_{R \succeq 0} \|A(H) + R\|_F^2.$$

Take a minimizer for G and restrict it to the principal block on U . The local assertion supplies a vertex whose full row/column contribution exceeds $17/16$; deleting it and using the remaining principal submatrix as a competitor for $G - v$ proves (??). $\square$

2 The parity reduction

A theta graph is bipartite exactly when a, b, c have the same parity; then (??) gives $s^+ = m = n + 1$.

Suppose it is nonbipartite. Relabel so that $a \equiv b \pmod{2}$ and c has the opposite parity. If $c \geq 4$, choose three consecutive internal vertices on the c -path. They induce a P_3 . Deleting any of the three breaks the c -path and leaves a connected bipartite unicyclic graph: its unique cycle has length $a + b$, which is even. Every possible deletion has $n - 1$ edges, so (??) and Lemma ?? give

$$s^+(G) \geq n - 1 + \frac{17}{16} = n + \frac{1}{16}. \quad (7)$$

It remains to treat

$$\Theta(2r, 2s, 1) \ (r, s \geq 1), \quad \Theta(2r + 1, 2s + 1, 2) \ (r, s \geq 0, \ r + s \geq 1),$$

and $\Theta(2r, 2s, 3)$ with $r, s \geq 1$.

3 A bordered-cycle witness

We first settle the middle family. Let A be the adjacency matrix of an even cycle C_N , let u, v lie in different bipartition classes, and put $b = e_u + e_v$. Adding a common neighbor gives

$$M = \begin{pmatrix} A & b \\ b^T & 0 \end{pmatrix}.$$

Write $P = A_+$ for the positive spectral part, and put $p = b^T P b$ and $q = b^T P^2 b$. The matrix

$$X = \begin{pmatrix} P & Pb/2 \\ b^T P/2 & p/4 \end{pmatrix} \succeq 0$$

has a Gram factorization with block rows $P^{1/2}$ and $b^T P^{1/2}/2$. Substitution in (??) gives

$$s^+(M) \geq N + 2p - \frac{q}{2} - \frac{p^2}{16} \geq N + p - \frac{p^2}{16},$$

because $0 \preceq P \preceq 2I$ implies $q \leq 2p$.

For $N \geq 6$, the second and fourth cycle spectral moments are 2 and 6; the latter counts the six closed four-step walks from a cycle vertex. Hölder interpolation gives

$$\frac{1}{N} \operatorname{tr} |A| \geq \frac{2}{\sqrt{3}}.$$

Since u, v are in opposite classes, $|A|_{uv} = 0$, and

$$p = \frac{1}{N} \operatorname{tr} |A| + \mathbf{1}_{\{u \sim v\}} \geq \frac{2}{\sqrt{3}}.$$

Cauchy–Schwarz gives $N^{-1} \operatorname{tr} |A| \leq \sqrt{2}$, so $p < 3$. As $x - x^2/16$ increases on $[0, 3]$,

$$p - \frac{p^2}{16} \geq \frac{2}{\sqrt{3}} - \frac{1}{12} > 1.$$

In C_4 every opposite-class pair is adjacent, so $p = 2$ and the estimate is stronger. Therefore

$$s^+(\Theta(2r + 1, 2s + 1, 2)) > |V|. \quad (8)$$

4 The chord family

Write $\Theta(2r, 2s, 1)$ as an even cycle C_N plus a chord between same-class vertices u, v . Put

$$a = \frac{e_u + e_v}{\sqrt{2}}, \quad b = \frac{e_u - e_v}{\sqrt{2}}, \quad E = aa^T - bb^T,$$

and, for $P = A_+$,

$$x = a^T P a, \quad y = b^T P b, \quad q = b^T P^2 b.$$

The dihedral reflection interchanging u, v commutes with P , fixes a , and negates b ; hence $a^T P b = 0$. With

$$T = I - \frac{1}{3}bb^T, \quad \alpha = (1 - x)_+,$$

the matrix $X = TPT + \alpha aa^T$ is positive semidefinite. Exact expansion in (??) yields

$$s^+(A + E) - N \geq 2x - \frac{8}{9}y - \frac{2}{9}q - \frac{25}{81}y^2 + (1 - x)_+^2. \quad (9)$$

Bipartite symmetry and spectral Cauchy–Schwarz give $y^2 \leq q/2$. Indeed, if Π_+ is the positive spectral projector, then

$$y^2 \leq (b^T \Pi_+ b)(b^T P^2 b), \quad b^T \Pi_+ b \leq \frac{1}{2},$$

because the bipartition involution exchanges positive and negative spectral subspaces. On one bipartition class P^2 has quadratic form $A^2/2$, so

$$q = \frac{1}{2}b^T A^2 b = 1 - \frac{(A^2)_{uv}}{2} \leq 1.$$

Finally $a^T A a = 0$ and $|A| \geq A^2/2$ give

$$x = \frac{1}{2}a^T |A| a \geq \frac{1}{4}a^T A^2 a = \frac{2 + (A^2)_{uv}}{4} \geq \frac{1}{2}.$$

Thus $y \leq 1/\sqrt{2}$, $y^2 \leq 1/2$, and $2x + (1 - x)_+^2 \geq 5/4$. Hence the right side of (??) is at least

$$\frac{5}{4} - \frac{8}{9\sqrt{2}} - \frac{2}{9} - \frac{25}{162} = \frac{283}{324} - \frac{4\sqrt{2}}{9} > 0.$$

Thus

$$s^+(\Theta(2r, 2s, 1)) > |V|. \quad (10)$$

5 The length-three ear

Let $A_0 = A(C_N) \oplus A(K_2)$, and let B be the symmetric adjacency matrix of the two matching edges joining the endpoints of K_2 to same-class cycle vertices u, v . The theta adjacency matrix is $A = A_0 + B$. Put $X_0 = (A_0)_+$ and $S = \text{tr}(X_0^2) = N + 1$. For $t \geq 0$ let

$$X_t = (I + tB)X_0(I + tB) \succeq 0.$$

Scaling a fixed witness in (??) gives, provided $\text{tr}(AX_t) > 0$,

$$s^+(A) \geq \frac{\text{tr}(AX_t)^2}{\text{tr}(X_t^2)}. \quad (11)$$

By vertex transitivity write $a = (A(C_N)_+)_{uu} = (A(C_N)_+)_{vv}$ and $b = (A(C_N)_+)_{uv}$. Block multiplication gives

$$\text{tr}(AX_t) = S + (2 + 4a)t + 2bt^2, \quad (12)$$

$$\text{tr}(X_t^2) = S + (6 + 8a + 8b)t^2 + (1 + 2a^2 + 2b^2)t^4. \quad (13)$$

We have $1/2 \leq a \leq 1/\sqrt{2}$ and $|b| \leq a$: use $a = |A(C_N)|_{uu}/2$, the inequalities $A(C_N)^2/2 \preceq |A(C_N)|$ and $|A(C_N)|_{uu} \leq \sqrt{(A(C_N)^2)_{uu}} = \sqrt{2}$, and the $\{u, v\}$ principal minor of $P \succeq 0$.

Take $t = 1/3$; positivity of $\text{tr}(AX_t)$ follows from (??) and $|b| \leq a$. Let U, D denote the quantities in (??)–(??). Direct expansion of $E = U^2 - (S + 1)D$ gives

$$\begin{aligned} 81E = & -2Sa^2 + 144Sa - 2Sb^2 - 36Sb - 28S \\ & + 142a^2 + 48ab + 72a + 2b^2 - 48b - 19. \end{aligned}$$

The coefficient of S is at least $-4a^2 + 108a - 28 \geq 25$, so this increases with $S \geq 5$. At $S = 5$ it is at least

$$132a^2 + 48ab + 792a - 8b^2 - 228b - 159.$$

Its derivative in b is $48a - 16b - 228 < 0$, so it decreases on $[-a, a]$. At $b = a$ it is $172a^2 + 564a - 159 \geq 166$. Thus $E > 0$, and (??) proves

$$s^+(\Theta(2r, 2s, 3)) > N + 2 = |V|. \quad (14)$$

Equations (??), (??), (??), and (??), together with the bipartite case, prove Theorem ??.

6 Verification

The source repository contains the complete attack notebook, detailed proof notes, and the exact script `experiments/theta_paper_certificate.py` checking the Bernstein coefficients and displayed trace identities. These computations are supplementary; the proofs above do not depend on finite numerical verification.

References

- [1] S. Akbari, H. Kumar, B. Mohar, S. Pragada, and S. Zhang, *Refinement of a conjecture on positive square energy of graphs*, arXiv:2506.07264v1, 2025.
- [2] C. Elphick, M. Farber, F. Goldberg, and P. Wocjan, *Conjectured bounds for the sum of squares of positive eigenvalues of a graph*, Discrete Math. 339 (2016), 2215–2223.
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